

BreakoutICU

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```
library(rpart)
library(rpart.plot)
library(rattle)
```

```
## Loading required package: tibble
```

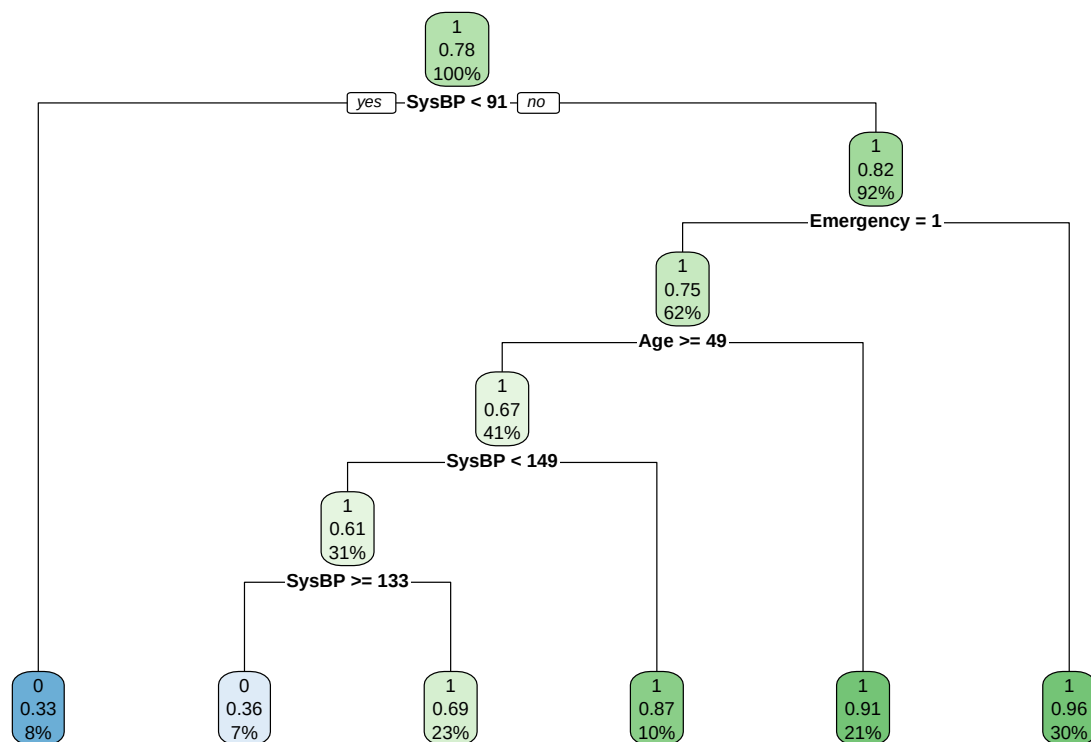
```
## Loading required package: bitops
```

```
## Rattle: A free graphical interface for data science with R.
## Version 5.5.1 Copyright (c) 2006-2021 Togaware Pty Ltd.
## Type 'rattle()' to shake, rattle, and roll your data.
```

```
icu_data = read.csv("ICU.csv")
icu_train = icu_data[1:150,]
icu_test = icu_data[151:200,]
attach(icu_train)

my_tree = rpart(formula = Survive ~ Age + AgeGroup + Sex + Infection + SysBP +
                 Pulse + Emergency, data = icu_train, method = "class", cp = 0.015)

rpart.plot(my_tree)
```



Those with systolic bloodpressure below 91 have a lower survival rate (33%) than those with higher than 91 (82%). Additionally, patients who didnt consulted the emergency room prior to thier ICU visit had a 96% survival rate, whereas those who did, ex. for a gunshot wound, car crash, or sudden event, had a 75% rate.

```
my_prediction = predict(my_tree, icu_test, type = "class")
table(my_prediction, icu_test$Survive)
```

```
##
## my_prediction  0  1
##              0  4  6
##              1  3 37
```

```
mean(my_prediction == icu_test$Survive)
```

```
## [1] 0.82
```

Here we can see that the 0.82 represents the accuracy of our model: 82% correct.

```
library(randomForest)
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

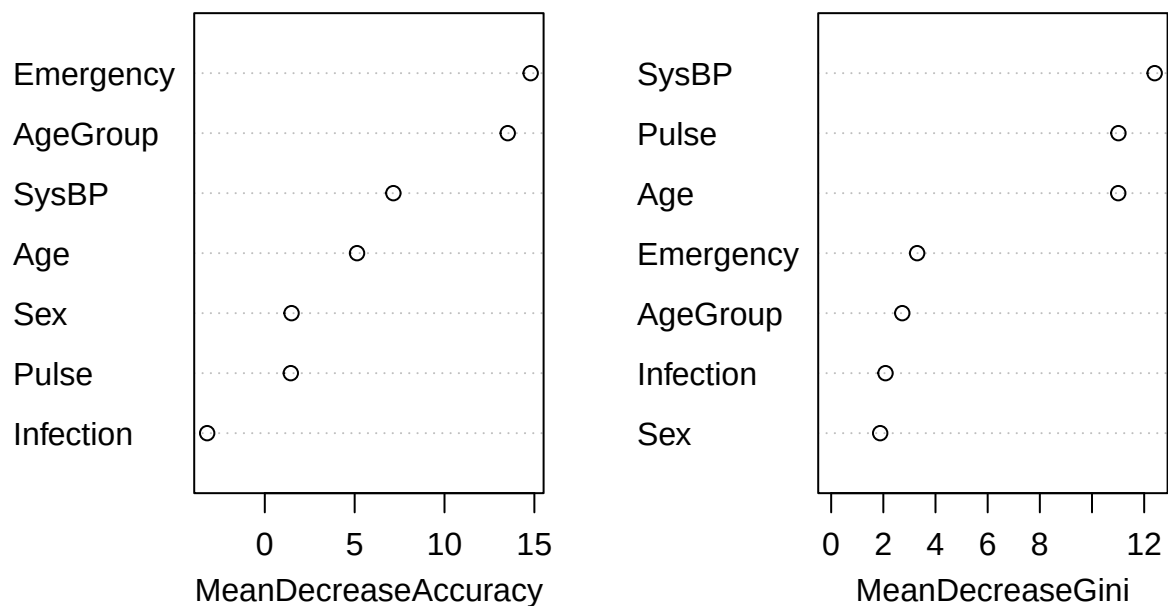
```
##
## Attaching package: 'randomForest'

## The following object is masked from 'package:rattle':
##
##      importance

my_forest <- randomForest(factor(Survive) ~ Age + AgeGroup + Sex + Infection + SysBP
                           + Pulse + Emergency, data= icu_train,
                           importance = TRUE, ntree = 1000)

varImpPlot(my_forest)
```

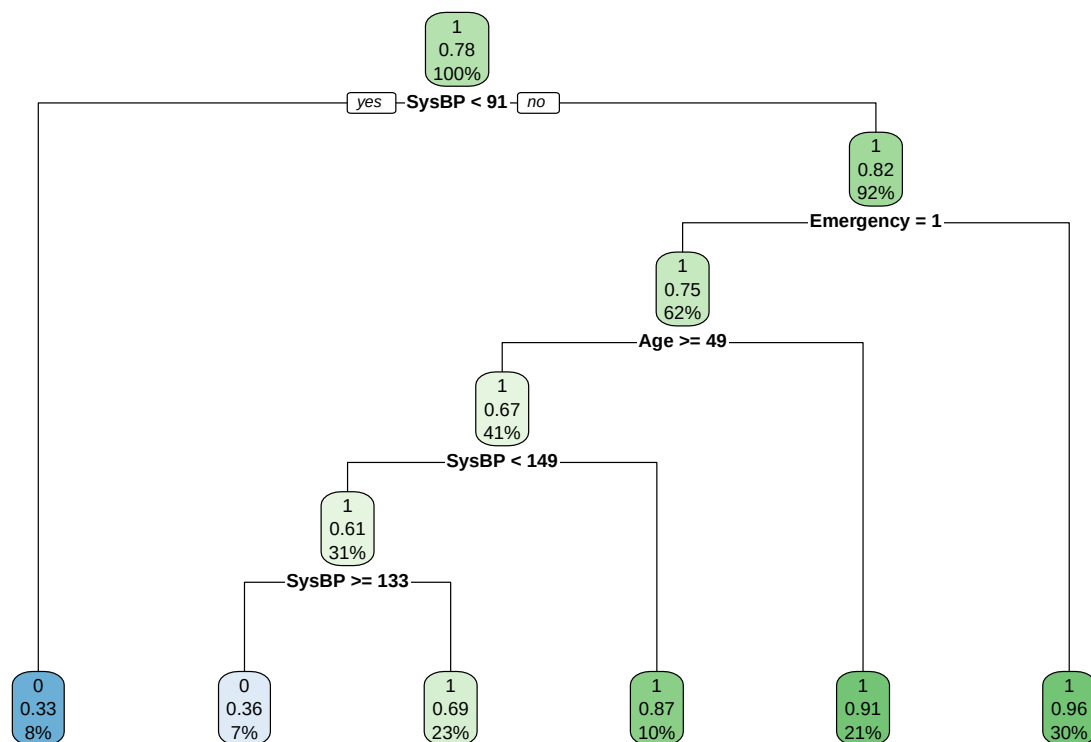
my_forest



By focusing on the MeanDecreaseAccuracy plot, the Infection predictor stands out as a variable of least importance, meaning that its removal from the model could improve efficiency if it truly has no impact on our accuracy result. We test this below.

```
my_tree2 = rpart(formula = Survive ~ Age + AgeGroup + Sex + SysBP +
                  Pulse + Emergency, data = icu_train, method = "class", cp = 0.015)

rpart.plot(my_tree2)
```



```
my_prediction2 = predict(my_tree2, icu_test, type = "class")
table(my_prediction2, icu_test$Survive)
```

```
##
## my_prediction2  0  1
##               0  4  6
##               1  3 37
```

```
mean(my_prediction2 == icu_test$Survive)
```

```
## [1] 0.82
```

Because there was no change in %accuracy between the first run and this run, the removal of predictor: “Infection” does not impact the data substantially, therefore its removal increases the efficiency of the program.